

Substance Use Disorder and Posttraumatic Stress Disorder Comorbidity: Addiction and Psychiatric Treatment Rates

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This study compares substance use disorder (SUD) patients with and without a comorbid diagnosis of posttraumatic stress disorder (PTSD) on their use of addiction and psychiatric services over the 6-month period before an inpatient substance abuse admission. Compared with non-PTSD patients, PTSD patients had a greater number of hospital overnights for addiction treatment. Given no significant between-groups differences on any substance use indexes, PTSD patients apparently overuse costly inpatient addiction services. Despite their greater rates of psychiatric comorbidity, PTSD patients did not receive treatment for psychiatric problems at greater rates than did non-PTSD patients. Among PTSD patients, use of PTSD treatment was low. Assessment of psychiatric comorbidity and referral to treatment targeting co-occurring PTSD and other disorders are suggested as possible ways to reduce the high treatment costs associated with SUD-PTSD comorbidity.

The presence of comorbid posttraumatic stress disorder (PTSD) has been associated with poorer substance use disorder (SUD) outcomes. For example, Brown, Stout, and Mueller (1996) compared substance-dependent women with and without a comorbid diagnosis of PTSD on their alcohol- and drug-use status 3 months postdischarge from inpatient substance abuse treatment. Although endpoint rates of relapse did not differ by PTSD status, women with PTSD were found to relapse more quickly than women who did not have PTSD. A proportional hazards regression showed that PTSD status was a significant predictor of time to relapse. In

another prospective study, Brown and Stout (1997) tracked 56 SUD patients (32 with PTSD and 24 without PTSD) for 6 months after their discharge from inpatient substance abuse treatment. They found that PTSD and non-PTSD substance-dependent patients did not differ in overall relapse rates or overall percentage of days abstinent. However, compared with non-PTSD patients, PTSD patients relapsed faster, drank more on those days when they did drink, and had more heavy drinking days during the follow-up period.

Although extant research shows that SUD-PTSD patients present a more severe and complex symptom picture (e.g., Brady, Killeen, Saladin, Dansky, & Becker, 1994), their poorer alcohol and drug outcomes appear specific to PTSD rather than to greater psychopathology in general (see Ouimette, Ahrens, Moos, & Finney, 1997). It has been speculated that SUD-PTSD patients fare worse than their non-PTSD counterparts because they do not receive adequate treatment for their PTSD and other associated problems. The underdiagnosis of PTSD in SUD patients (Dansky, Roitzsch, Brady, & Saladin, 1997), the tendency to classify anxiety and depression symptoms as secondary to the alcohol–drug problem (Kofoed, Friedman, &

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Peck, 1993), and both patients' and treatment staffs' discomfort with the topic of trauma and PTSD (Turner & Colao, 1985) all may contribute to PTSD patients not receiving needed psychiatric treatment.

To date, relatively little is known about the treatment-use patterns of SUD-PTSD patients. To our knowledge, only three studies have examined the SUD treatment histories of individuals suffering from both SUDs and PTSD. Using a large sample of treatment-seeking male veterans, Ouimette et al. (1997) found that at 1-year follow-up SUD-PTSD patients were more likely to be readmitted for addiction-mental health inpatient treatment compared with patients with SUD only. Brown, Recupero, and Stout (1995) studied male and female treatment-seeking SUD patients and found that those with possible PTSD reported a greater number of lifetime inpatient addiction admissions than those patients without PTSD. In another study, Druley and Pashko (1988) tracked male veterans' medical records after discharge from inpatient substance abuse treatment. They found that 70% of the PTSD patients' charts, compared with 30% of the non-PTSD groups', were on another unit, perhaps indicating that PTSD patients use hospital services more than their non-PTSD counterparts. Despite different methodologies and different sampling procedures, these three studies indicate that PTSD patients tend to use costly inpatient treatment for their addiction problems. However, given that these studies did not comprehensively assess other types of treatment (e.g., outpatient addiction and psychiatric treatment), we cannot conclude that PTSD patients overuse or rely exclusively on inpatient addiction services or that they use these services in lieu of needed psychiatric treatment.

The purpose of the present study is to assess and compare the rates of addiction and psychiatric treatments used over a 6-month period by SUD patients with and without PTSD. An examination of patients' treatment-use patterns may help identify possible overused and underused services and indicate needed changes in the treatment of SUD-PTSD patients. We investigate possible group differences in the use of inpatient care (addiction, psychiatric) and outpatient services (addiction, psychiatric). For inpatient addiction services, we estimate treatment

costs for PTSD versus non-PTSD substance-dependent patients. Treatment for trauma-PTSD is assessed separately from psychiatric treatment, and use rates are examined for SUD-PTSD patients.

In keeping with earlier research, we expect that SUD-PTSD patients will be more likely to use costly inpatient addiction treatment and have a significantly greater total number of nights in the hospital compared with non-PTSD patients. Despite expected between-group differences in psychiatric comorbidity (i.e., other non-PTSD diagnoses), we do not predict that PTSD patients will have used psychiatric treatment at a greater rate than their non-PTSD counterparts. Finally, among the PTSD group, rates of trauma-PTSD treatment will be low.

Method

Participants

Participants were 51 women and 44 men who were receiving inpatient substance abuse treatment at a freestanding, private, university-affiliated hospital. They ranged in age from 18 to 55 years, with a mean age of 37.60 ($SD = 8.79$). Approximately 88% of the sample were Caucasian, 10% were African-American and 2% were American Indian. Approximately 38% were separated or divorced; 37% were married or living with a partner; and 25% had never married. Roughly half the sample (47%) were unemployed or disabled, 44% were working full-time, and 9% were working part-time. Participants averaged slightly more than a high school education (13.03 years, $SD = 2.05$) and had a modal personal income in the \$0-\$9,999 range.

Approximately 41% of the sample carried a *Diagnostic and Statistical Manual of Mental Disorders* (4th ed.) (*DSM-IV*; American Psychiatric Association, 1994) diagnosis of alcohol dependence only, 28% were diagnosed with drug dependence only, and 31% were dependent on both alcohol and drug(s). Participants had abused substances for an average of 14 years ($SD = 10.19$). In the 180 days before their inpatient admission, patients reported being abstinent 48% of the time. A total of 92 collaterals (97%) provided data on their respective participant's percentage of days abstinent during the 180 days before admission. Agreement between participants' and collaterals' reports was fair (intra-class correlation coefficient = .47, $p < .01$). The majority of discrepancies consisted of collaterals reporting that participants had more days abstinent than the participants themselves reported, suggesting

that participants were forthright about their substance use during the follow-up period.

Procedures

Participants were recruited from an inpatient substance abuse treatment unit at a private psychiatric hospital. Treatment was multimodal (general psychotherapy, psychopharmacology, and detoxification) but included no specific therapy targeting trauma-PTSD. To be eligible for study participation, patients had to be literate, between the ages of 18 and 55 years, and able to provide the name and address of one collateral who could serve as a corroborative source. Exclusionary criteria included the presence of significant neurological or organic impairment, psychosis, and homelessness. A total of 161 patients were approached for study participation. Thirty of these patients did not meet our eligibility criteria; another 35 refused to participate (22 were not interested in participating in a research study, 9 cited health issues as an impediment to participation, and 4 were in the process of being discharged from the hospital). According to chart reviews, nonparticipants did not differ from participants on PTSD diagnosis. One woman who agreed to participate left the inpatient unit before completing the entire assessment. She had reported an extensive trauma history; however, no PTSD diagnostic information was obtained. Data were analyzed on the remaining 95 patients.

Measures

Health care use and costs. During their inpatient stay, participants completed an extensive health care interview, which assessed the type and amount of services used in the 6 months before their current inpatient admission. This assessment inquired about the number of hospital overnights (i.e., the total number of nights spent in the hospital) and number of outpatient visits both for alcohol-drug detoxification-treatment and for psychiatric treatment. Psychiatric treatment did not include treatment for trauma-PTSD, which was assessed separately. Average overnight costs for addiction treatment were estimated by means of 1994 figures available from the Agency for Health Care Policy and Research (1997). According to this source, the mean length of stay was 7.26 days and the mean total charge was \$5,148 (no standard errors reported), resulting in an average cost of \$709 per day.

SUDs, PTSD, and other psychiatric comorbidity. We measured substance use during the 180 days before inpatient treatment by means of the Time-Line Follow-Back (TLFB; Sobell, Maisto, Sobell, & Cooper, 1979). The TLFB was used to obtain the

following substance use indexes: percentage of days abstinent, drinks per drinking day, and percentage of heavy drinking days. A trained interviewer, skilled in making differential diagnoses, administered the Structured Clinical Interview for *DSM-IV* (SCID; Spitzer, Williams, Gibbon, & First, 1994) to assess for SUDs, affective disorders and all non-PTSD anxiety disorders. Given assessment time constraints, we were unable to administer the SCID for Axis II disorders. However, we did conduct a comprehensive review of chart diagnoses for any Cluster B personality disorder (e.g., borderline, antisocial, narcissistic, histrionic) and classified patients as present or absent on this Axis II dimension.

Participants also were administered the Life Stressor Checklist-Revised (Wolfe, Kimerling, Brown, Chrestman, & Levin, 1996), a 30-item screening measure of stressful events across the lifespan. Sample events include physical assault, sexual abuse-assault, being robbed or mugged, and the catastrophic death of a loved one. If patients reported experiencing a traumatic event as defined in *DSM-IV* (American Psychiatric Association, 1994), they were administered the Clinician Administered PTSD Scale (CAPS; Blake et al., 1995). The CAPS is a structured interview assessing the frequency and severity of the 17 cardinal symptoms of PTSD according to *DSM-IV* criteria. Participants also completed the Revised Symptom Checklist (SCL-90-R; Derogatis, 1977), a widely used, 90-item, self-report measure that yields an overall score indicative of general psychiatric distress.

Statistical Methods

We divided participants into those with and without a current diagnosis of PTSD. Comparisons between PTSD and non-PTSD groups were made for sociodemographic variables, TLFB substance use indexes, traumatic exposure, comorbidity, and types and rates of mental health care use. For categorical variables, chi-square tests were calculated (or Fisher's exact test when expected cell frequencies were small). *T* tests were calculated for continuous variables.

Results

Approximately half the sample ($n = 48$) met *DSM-IV* criteria for current PTSD. Almost all of the participants (95%, $n = 90$) reported that they had been exposed to some Criterion A event during their lifetime. Commonly reported traumas included catastrophic death of a loved one (e.g., homicide or suicide; 65%), childhood forced sexual touching (47%), childhood physical abuse (40%), and adulthood physical abuse

(38%). Women were more likely than men to have experienced adulthood forced sex (31% vs. 5%), $\chi^2(1, N = 95) = 11.07, p < .001$, and adulthood physical abuse (59% versus 14%), $\chi^2(1, N = 95) = 20.49, p < .0001$. Women also experienced more types of trauma ($M = 3.98, SD = 2.21$) than did men ($M = 2.84, SD = 1.72$), $t(92) = 2.82, p < .01$. Despite women's greater exposure to trauma in general and to adulthood physical and sexual abuse in particular, the rate of concurrent PTSD did not significantly differ by gender.

PTSD and non-PTSD groups did not significantly differ on any demographic characteristics. Of note, there also were no between-group differences on any substance use behaviors (number of years of problematic substance use, percentage of days abstinent, drinks per drinking day, percentage of days of heavy drinking, type of substance use disorder). Gender differences were found for number of years of problematic substance use: Men had misused alcohol and drugs for an average of 17 years ($SD = 10.04$), women had abused alcohol and drugs for an average of 11 years ($SD = 9.66$), $t(90) = 2.77, p < .01$.

Comorbidity (Other Than PTSD)

The majority of study participants (78%) carried another Axis I diagnosis other than PTSD or substance dependence. Approximately

two thirds of the sample had a current affective disorder; over half had some form of a current anxiety (non-PTSD) disorder. PTSD patients had a significantly greater number of other Axis I disorders ($M = 2.48, SD = 1.37$) than did their non-PTSD counterparts ($M = 1.27, SD = 1.27$), $t(93) = 4.06, p < .001$. Patients with PTSD were more likely to meet criteria for current major depressive disorder, bipolar disorder, and panic disorder (with or without agoraphobia). PTSD patients also were more likely to suffer from a Cluster B personality disorder (see Table 1). In keeping with the high rates of psychiatric comorbidity, PTSD patients reported significantly greater psychiatric distress (as measured by the SCL-90-R; $M = 168, SD = 58.49$) than did non-PTSD patients ($M = 110, SD = 63.44$), $t(92) = 4.64, p < .001$. Comorbidity rates did not differ by gender.

Treatment Use

The PTSD group was not significantly more likely than the non-PTSD group to have ever used addiction services (inpatient or outpatient) or psychiatric services (inpatient or outpatient) in the previous 180 days. However, as shown in Figure 1, PTSD patients had a significantly greater number of hospital overnight stays for substance-abuse-related treatment than did non-PTSD patients, $t(93) = 2.05, p < .05$. Given that the PTSD group's higher inpatient rates

Table 1
Rates of Comorbid Diagnoses in Presence or Absence of Current PTSD

Diagnoses	Total Ns	PTSD		Non-PTSD	
		n	%	n	%
Major depressive disorder ^a	62	38	79	24	51
Dysthymia	12	9	19	3	6
Mania ^b	7	7	15	0	0
Panic disorder ^c	32	23	48	9	19
Agoraphobia without panic disorder	1	0	0	1	2
Social phobia	14	8	17	6	13
Simple phobia	10	3	6	1	2
Obsessive-compulsive disorder	10	7	15	3	6
Generalized anxiety disorder	17	9	19	8	17
Cluster B personality disorder ^d	20	15	31	5	11

Note. For PTSD group, $n = 48$; for non-PTSD group, $n = 47$. PTSD = posttraumatic stress disorder.
^a $\chi^2(1, N = 62) = 8.27, p = .00402$. ^b $\chi^2(1, N = 7) = 7.40, p = .00652$. ^c $\chi^2(1, N = 32) = 8.79, p = .00302$. ^d $\chi^2(1, N = 20) = 6.07, p = .01375$.

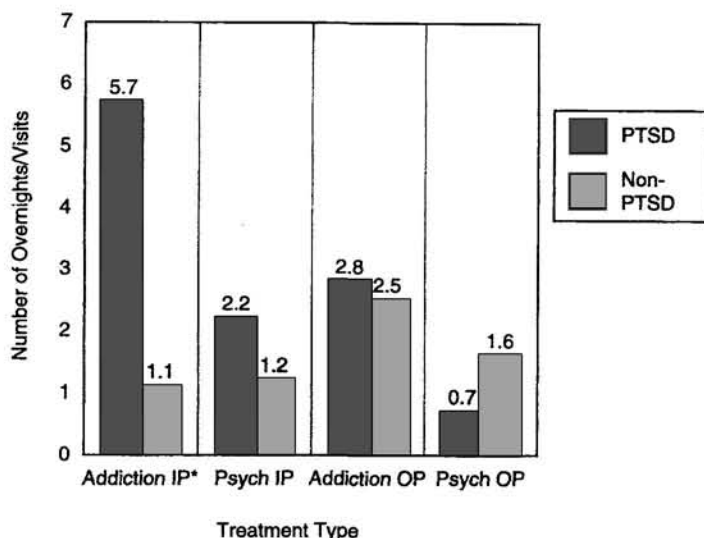


Figure 1. Six-month treatment-utilization rates by posttraumatic stress disorder (PTSD) status. IP = inpatient; OP = outpatient. $*t(93) = 2.05, p = .043$.

could be attributable to their greater comorbidity in general rather than to PTSD in particular, we conducted an analysis of covariance with additional comorbidity (defined as the total number of disorders other than PTSD and SUDs) as the covariate, number of addiction hospital overnight stays as the dependent variable, and PTSD as the independent variable. This analysis was significant ($F = 31.13, p < .001$), indicating that PTSD remained a significant factor even after controlling for additional comorbidity.

No significant between-group differences were found for the total number of psychiatric overnight stays or number of outpatient visits (substance abuse or psychiatric) during the previous 180 days. An examination of treatment use (type and amount) for men versus women revealed no significant gender differences.

Of the 48 patients with PTSD, only 14 (29%) had received any type of trauma treatment in the previous 6 months. Two patients had received inpatient treatment specifically for their PTSD (one for 5 overnight stays and the other for 10 overnight stays). All 14 patients had received outpatient PTSD treatment. The number of outpatient PTSD visits ranged from 1 to 20, with a mean of 2.54 visits ($SD = 4.86$).

Inpatient Addiction-Treatment Costs

Using the estimate of \$709 per overnight addiction visit (Agency for Health Care Policy and Research, 1997), we found that PTSD patients incurred an average cost of \$4,042 in a 6-month period versus \$780 for non-PTSD patients. Relative to non-PTSD patients, PTSD patients incurred an additional \$3,262 for inpatient addiction treatment in the span of 6 months.

Discussion

In keeping with previous research with community samples (e.g., Kessler et al., 1996) and clinical samples (e.g., Brady et al., 1994, Brown et al., 1996), we found high rates of SUD-PTSD comorbidity. Compared with their non-PTSD counterparts, SUD-PTSD patients evidenced higher levels of self-reported psychiatric distress and a greater likelihood of meeting diagnostic criteria for current major depressive disorder, bipolar disorder, panic disorder, and a Cluster B personality disorder. Despite their greater rates of psychiatric comorbidity, PTSD patients had not used inpatient or outpatient treatment for psychiatric problems (excluding

PTSD-specific treatment) more than non-PTSD patients had. Only 1 in 4 of the PTSD patients had received any type of non-PTSD psychiatric treatment, averaging one outpatient visit and two hospital overnight stays. Although it was beyond the scope of this study to determine the reason for these low psychiatric-treatment-utilization rates, we would speculate that SUD-PTSD patients typically were not referred to appropriate psychiatric treatment. Because study participants were selected from an inpatient addiction unit, either the patient or the treatment system apparently identified SUD as the sole or dominant disorder. Given the overlap between anxiety and depressive symptoms and SUDs (particularly during withdrawal; Weiss, Mirin, & Griffin, 1992), it might be difficult to detect comorbid problems. However, our findings indicated that substance abuse treatment providers need to screen for possible independent and potentially complicating conditions. Without such diagnostic assessments and subsequent treatment referrals, PTSD patients might not receive adequate care for their psychiatric problems, which might ultimately lead to poor treatment outcomes.

Similar to rates of psychiatric care, use of PTSD treatment was low. Only 1 in 4 PTSD patients had received PTSD treatment (primarily outpatient) in the previous 6 months. Research on 42 of our SUD-PTSD patients (see Brown, Stout, & Gannon-Rowley, 1998) indicates that this low treatment rate is attributable to treatment providers' lack of referral rather than patients' lack of follow-through or noncompliance. These findings suggest the need to educate treatment staff about PTSD, its connection with SUDs, and possible treatment options. Without guidance and referral from treatment providers, it is unlikely that SUD-PTSD patients will receive PTSD treatment.

Although SUD-PTSD patients were not more likely to have ever used inpatient alcohol-drug treatment in the past 6 months, they did have a greater number of hospital overnight stays for addiction treatment compared with their non-PTSD counterparts. Given that PTSD and non-PTSD patients did not significantly differ in their substance use behaviors (e.g., percentage of days abstinent), their greater use of inpatient addiction treatment cannot be attributed to more severe or chronic substance use problems. The

PTSD group's greater number of hospital overnight stays for SUD problems also cannot be explained by their greater psychiatric comorbidity or by differential insurance coverage for addiction versus psychiatric inpatient treatment. PTSD patients apparently overutilize expensive inpatient addiction services relative to non-PTSD patients. Moreover, PTSD patients appear to underutilize needed psychiatric treatment. A reallocation of treatment services may be beneficial for SUD-PTSD patients and the programs and staff serving them. Greater use of treatment targeting comorbid PTSD and other mental disorders might result in improved substance abuse outcomes. A more balanced overall treatment plan may ultimately relieve the excessive burden on inpatient SUD treatment programs and reduce treatment costs associated with SUD-PTSD comorbidity.

Using the most recent figures available on inpatient addiction-treatment costs (Agency for Health Care Policy and Research, 1997), our treatment-utilization data indicate that PTSD patients incurred an additional \$3,262 for inpatient addiction-treatment costs relative to other patients. Although we do not know the extent to which these differences in inpatient addiction-service use can be generalized across time, the existence and magnitude of the difference over a 6-month span are noteworthy. Clearly, PTSD patients constitute an important target for focused interventions to reduce health care expenditures.

There are several study limitations that are important to note. The nature of the sample, selected while receiving inpatient addiction treatment, may limit the generalizability of the findings. SUD patients seeking inpatient care may be more impaired and not representative of those patients who seek alternative forms of addiction services. Although our sample reflects the racial distribution of the adult population in Rhode Island, the low percentage of minorities precluded any analyses by racial status. Future research could replicate our study with a more racially diverse sample of SUD patients. Another limitation is that we assessed patients near the time of their detoxification, which possibly inflated our comorbidity rates. However, we diagnosed additional disorders only if we were certain they were independent of SUDs and consequently feel confident in our differential

diagnostic assessments. Our use of chart diagnoses to estimate the prevalence of Cluster B personality disorders raises concerns about their validity and reliability. We recommend that future studies use a structured clinical interview in diagnosing comorbid Axis II disorders.

Regarding our inpatient addiction costs, we must caution that our estimate of overnight costs does not represent a true cost but rather an upper bound. The mean total charge reported by the Health Care Cost and Utilization Project (Agency for Health Care Policy and Research, 1997) was calculated before discounts and did not include copayments and deductibles. However, given that charges were based on 1994 data and the lack of real-cost data (U.S. Department of Health and Human Services, 1997), we think our estimates are acceptable. Note also that our 6-month utilization data were based solely on patients' self-reports. Although we verified patients' inpatient treatment received at the study's hospital site, we were unable to do so for services received at other treatment facilities.

Finally, the present study focused exclusively on inpatient and outpatient treatment for alcohol and drugs, PTSD, and other psychiatric problems. Future research warrants a broader assessment of mental health treatment usage, with particular attention paid to partial-day hospital programs, which are increasingly replacing costly inpatient programs (Sherman, 1992), and pharmacological treatments, which are showing promising results (e.g., sertraline; see Brady, Sonne, & Roberts, 1995). Given research showing that SUD-PTSD patients' outcomes are enhanced by their involvement in 12-step activities (Ouimette, Ahrens, Moos, & Finney, 1998), assessment of utilization rates of such inexpensive and readily available self-help treatment also would be beneficial. Such research might help further elucidate ways to improve outcomes for SUD-PTSD patients and reduce the high costs of treating addiction in the presence of PTSD.

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Summary of Journal Operations for 1998

The summary below is compiled from manuscripts received in 1998.

Manuscripts		Printed pages	
No. pending 1/1/98 ^a	30	Items published ^f	23
No. new items received ^b	101	Total pages published ^g	281
No. new manuscripts received ^c	99		
No. accepted ^d	28		
No. pending 12/31/98 ^e	35		
Rejection rate	71%		

^aIncludes manuscripts for which final editorial decisions had not been made as of January 1, 1998.

^bIncludes all manuscripts, special series introductions, reports, and book reviews received.

^cIncludes manuscripts and book reviews only.

^dIncludes 20 manuscripts submitted before 1998 that were revised, resubmitted, and accepted in 1998.

^eIncludes manuscripts for which final editorial decisions had not been made as of December 31, 1998.

^fIncludes articles, reports, comments, and reviews.

^gIncludes articles, tables of contents, and author index.